Fair Territory Division over Discrete Postal Codes

Formulation, algorithm, and a worked fifty-zip instance

Abstract

A companion to the continuum notes, restating the fair-division problem directly on postal codes. Ordering zips by utility ratio reduces the utilitarian and Nash problems to choosing one index among $n+1$, evaluable in a single pass; the equalisation criteria do not inherit that structure, and we show where the prefix restriction breaks. The mixture-quantile shortcut, exact on a continuum, is off by one zip on discrete data. Contiguity is then imposed exactly by a districting integer program, which turns out to cost far less than a greedy repair suggests, provided the objective carries both a compactness term and a welfare floor. A fifty-zip instance drawn from a Gaussian mixture is worked through in full: distributions, solution, opportunity balance, contestability, and the exact parameter breakpoints. Headroom is net throughout.

1 Setup

Index postal codes $z = 1, \ldots, n$. For each, three observed quantities: sales A_z by wholesaler a, sales B_z by wholesaler b, and total market opportunity M_z. Totals are $S_a = \sum_z A_z$, $S_b = \sum_z B_z$, $M = \sum_z M_z$. With transfer capture θ and headroom credit λ, and headroom taken *net*,

$$c_1 = 1 - \lambda, \qquad c_2 = \theta(1-\lambda), \qquad \Lambda = c_1 + c_2 = (1-\lambda)(1+\theta), \tag{1}$$

$$u_a(z) = c_1 A_z + c_2 B_z + \lambda M_z, \qquad u_b(z) = c_2 A_z + c_1 B_z + \lambda M_z. \tag{2}$$

Headroom must be non-negative pointwise: $M_z \geq A_z + \theta B_z$ and $M_z \geq B_z + \theta A_z$.

An allocation is a set $S \subseteq \{1, \ldots, n\}$ given to a, with the complement to b. Gains against pre-merger production are

$$g_a(S) = \sum_{z \in S} u_a(z) - S_a, \qquad g_b(S) = \sum_{z \notin S} u_b(z) - S_b, \tag{3}$$

and the attainable maxima, from receiving every zip, are $G_a^{\max} = \sum_z u_a(z) - S_a$ and $G_b^{\max} = \sum_z u_b(z) - S_b$.

2 The utility ratio orders everything

Define

$$r_z = \frac{u_a(z)}{u_b(z)}. \tag{4}$$

Moving zip z from b to a raises g_a by $u_a(z)$ and lowers g_b by $u_b(z)$, so r_z is the exchange rate at which utility transfers between the two wholesalers.

Proposition 1 (Prefix property)**.** *Order the zips so that $r_{(1)} \geq \cdots \geq r_{(n)}$ and write $P_k = \{(1), \ldots, (k)\}$. Then:*

(i) the ***utilitarian*** *optimum is exactly P_k with $k = |\{z : r_z > 1\}|$;*

(ii) the ***Nash*** *optimum is some P_k, with threshold $r = g_a/g_b$.*

For (i), the objective is $\sum_{z \in S}\big(u_a(z) - u_b(z)\big) + \text{const}$, maximised by taking every zip with $u_a > u_b$. For (ii), at an optimum of $\log g_a + \log g_b$ every included zip satisfies $u_a(z)/g_a \geq u_b(z)/g_b$ and every excluded one the reverse, which is a threshold on r_z.

The prefix property does *not* extend to the equalisation criteria. Equal gain, Kalai–Smorodinsky and the egalitarian criterion all minimise a *distance* rather than maximise a monotone quantity, and an off-frontier subset can hit the target more precisely. Over 400,000 random subsets of the worked instance:

criterion	best prefix	best random subset	
utilitarian max $g_a + g_b$	32.983	32.894	prefix holds
Nash max $g_a g_b$	210.253	201.896	prefix holds
egalitarian max $\min(g)$	13.382	14.096	**violated**
equal gain min $\lvert g_a - g_b\rvert$	2.3287	0.0001	**violated**
Kalai–Smorodinsky	0.02240	0.000004	**violated**

The prefix table therefore remains exact for utilitarian and Nash, and is a fast, well-behaved *restriction* for the others — one that has the merit of staying on a family of allocations ordered by exchange rate. Exact optima for the equalisation criteria require search, which is what Section 9 provides.

3 The mixture-quantile shortcut, and why it fails here

On a continuum the equal-gain condition collapses to a statement about a mixture measure. Collecting (3) under $g_a = g_b$ gives

$$\Phi(S) = q, \qquad \Phi(S) = \frac{\Lambda \sum_{z \in S}(A_z + B_z) + 2\lambda \sum_{z \in S} M_z}{D}, \qquad q = \frac{S_a(1 + c_2) - \lambda S_b + \lambda M}{D}, \tag{5}$$

with $D = \Lambda(S_a + S_b) + 2\lambda M$. Kalai–Smorodinsky takes the same form with weights $K_a = S_a(G_b^{\max} c_1 + G_a^{\max} c_2)$, $K_b = S_b(G_b^{\max} c_2 + G_a^{\max} c_1)$, $K_M = \lambda M(G_a^{\max} + G_b^{\max})$ and level $q_{KS} = (G_a^{\max} G_b^{\max} + G_b^{\max} S_a)/K$.

This suggests an appealing shortcut: accumulate mixture mass along the ratio order and cut where it crosses q. On a continuum that is exact.

It is off by one zip on discrete data. The level q is a continuum target, while the cumulative mass *jumps* at each zip, so the crossing generally falls strictly between two prefixes and truncating downward need not give the nearer one. On the worked instance, with $q_{KS} = 0.523326$:

k	cumulative mass	distance to q_{KS}	$\lvert g_a/G_a^{\max} - g_b/G_b^{\max}\rvert$
23	0.511418	-0.011908	0.02864
24	0.517624	-0.005702	0.01371
25	0.535206	$+0.011881$	0.02857
26	0.544357	$+0.021031$	0.05058

Here truncation happens to give the better Kalai–Smorodinsky answer. On the *equal-gain* criterion the same truncation gives $k = 24$ with a gap of 0.449 where $k = 25$ gives 0.055 — eight times better. Whether truncation lands on the nearer integer is incidental.

The mixture form remains the right theory: it is what makes q interpretable, what yields closed-form comparative statics, and what reveals that λ governs the weight placed on opportunity (Section 5). It is the wrong *algorithm* for discrete data. Use Proposition 1 instead.

Remark (Nash needs no fixed point here). On a continuum the Nash threshold satisfies $t^\star = g_a/g_b$, a fixed point whose naive iteration diverges — on this instance it runs $1.0000 \to 40.22 \to -0.244 \to -6.447$ and locks into a degenerate corner. Discretely the question does not arise: enumerate the $n+1$ prefixes and take the largest feasible product. Forty-two of the fifty-one prefixes had both gains positive.

4 A fifty-zip instance

Fifty postal codes on the unit square, drawn from a two-component Gaussian mixture representing metropolitan areas at $(0.28, 0.62)$ and $(0.72, 0.36)$. Sales are generated so that a is anchored on the first metro and b on the second, with both present in each. Totals are set to $S_a = 3.0$, $S_b = 1.8$, $M = 40$, giving combined market penetration of 12%. Parameters $\theta = 0.4$, $\lambda = 0.3$.

combined penetration $(S_a + S_b)/M$	0.1200
$\mathrm{corr}(A_z, B_z)$ across zips	+0.685
utility ratio range	[0.9531, 1.1373]
minimum net headroom	0.2335

The positive correlation matters: both wholesalers are strong in the same places, which is the harder configuration. The ratio spans a narrow band and lies mostly above 1, so on raw utility a is favoured nearly everywhere and fairness is the only force conceding territory to b.

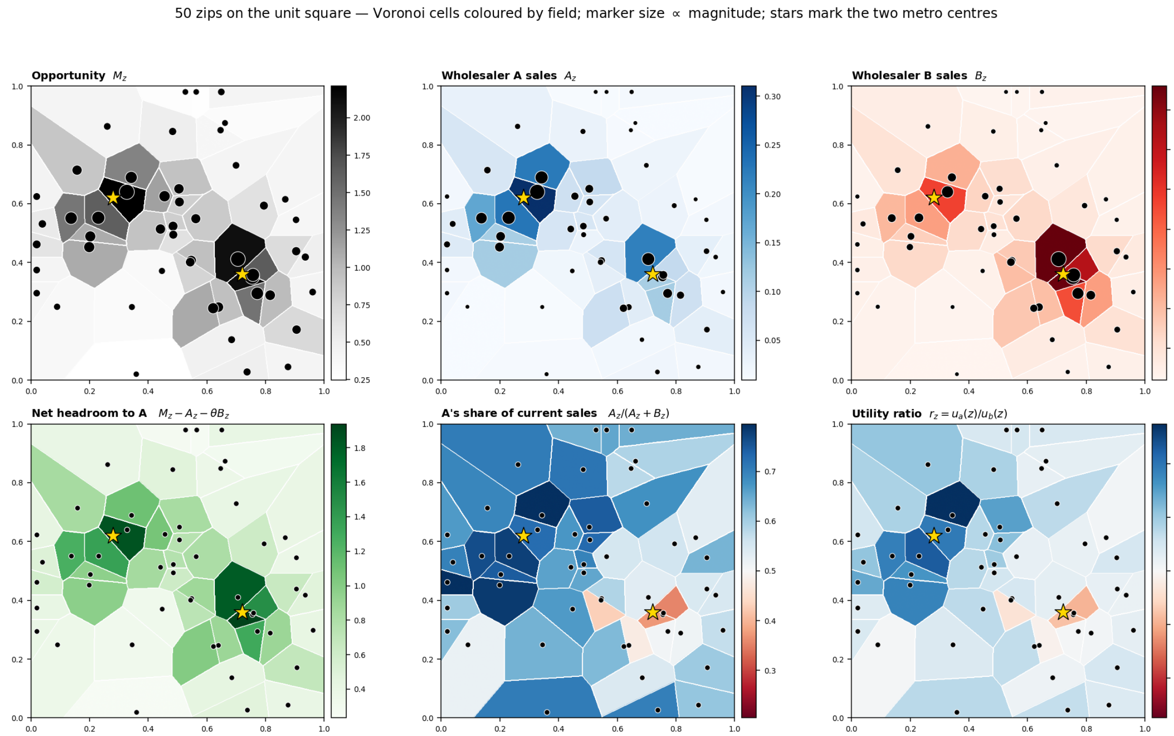

Figure 1: The instance before any optimisation. Voronoi cells coloured by field; marker size proportional to magnitude; stars mark metro centres. Opportunity concentrates in two metros; a's book is anchored north-west and b's south-east; net headroom is largest where opportunity is largest.

4.1 Solution

$G_a^{\max}$, $G_b^{\max}$	11.604, 12.300
KS mixture weights (F_a, F_b, F_M)	(0.1036, 0.0607, 0.8357)
q_{KS}	0.523326

Selecting each criterion directly over the fifty-one prefixes:

criterion	k	g_a	g_b	$g_a g_b$	$\min(g)$	$\lvert g_a/G_a^{\max} - g_b/G_b^{\max}\rvert$
Nash	25	4.9360	4.8807	**24.091**	4.8807	0.0286
equal gain	25	4.9360	4.8807	24.091	4.8807	0.0286
egalitarian	25	4.9360	4.8807	24.091	4.8807	0.0286
Kalai–Smorodinsky	24	4.6807	5.1301	24.012	4.6807	0.0137
equal opportunity	22	4.4423	5.3628	23.823	4.4423	0.0532
utilitarian	46	9.6330	0.2395	2.307	0.2395	0.8107

Nash, equal gain and the egalitarian criterion coincide at $k = 25$. Kalai–Smorodinsky genuinely prefers $k = 24$, because it equalises *fractions* of attainable gain and the maxima differ (11.60 against 12.30); that disagreement is substantive, not an artefact. All four allocations satisfy EF1, so envy-freeness does not discriminate between them. The utilitarian row shows what fairness is buying: unconstrained, a takes 46 zips and leaves b a gain of 0.24.

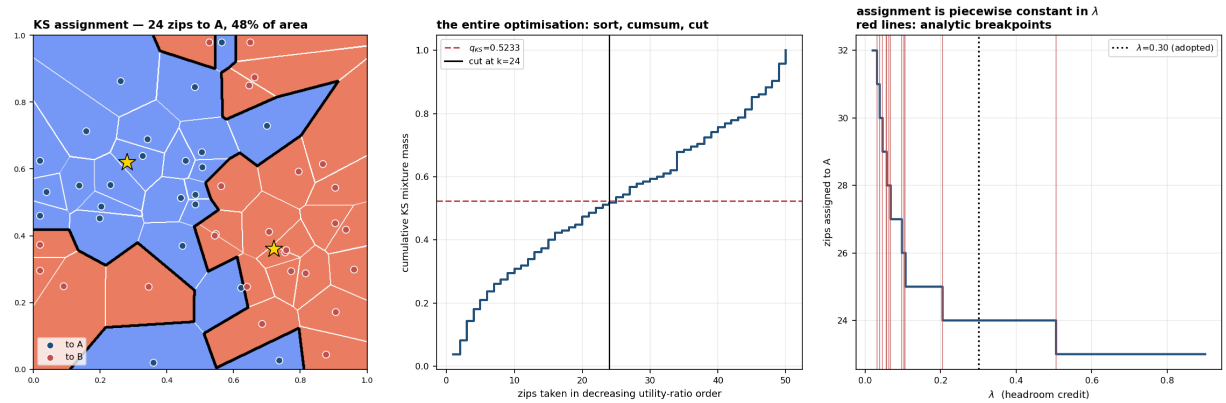

Figure 2: Left: the Kalai–Smorodinsky allocation, 24 zips to a covering 48% of the area and 52.6% of the value. Centre: the entire optimisation — sort, cumulative sum, cut. Right: the allocation is piecewise constant in λ, with breakpoints matching the closed-form prediction of Section 7.

Spatial coherence of the allocation, measured as six-nearest-neighbour label agreement, is 0.697: each wholesaler holds isolated pockets inside the other's region. That fragmentation is the contiguity cost made concrete, and is the natural next constraint to impose on the real adjacency graph.

5 Opportunity balance

Before the merger both wholesalers covered the entire market, so each had access to opportunity $M = 40$ at penetrations of 7.5% and 4.5%. The honest framing of the exercise is therefore that both end with roughly half of what they had.

	M_a	M_b	imbalance
before (full overlap)	40.000 (100%)	40.000 (100%)	—
Kalai–Smorodinsky	20.492 (51.2%)	19.508 (48.8%)	2.5%
equal opportunity	19.775 (49.4%)	20.225 (50.6%)	1.1%

Equal opportunity is simply another (measure, level) pair in the same algorithm: measure M_z/M, level 0.5. It moves two zips, halves the imbalance, and quadruples the KS fairness gap from 0.0137 to 0.0532.

λ is the opportunity-balance dial. The KS mixture already places 83.6% of its weight on the opportunity measure at $\lambda = 0.30$, so the criterion is largely balancing opportunity without being asked to.

λ	weight on F_M	M_a share	opportunity imbalance	$\lvert$KS gap$\rvert$
0.02	0.193	0.598	19.5%	0.0801
0.10	0.568	0.533	6.6%	0.0295
0.30	0.836	0.512	2.5%	0.0137
0.70	0.965	0.506	1.1%	0.0094

"How much should untapped opportunity count?" and "should territories carry equal market size?" are therefore the same question. Note also that with books of 3.0 against 1.8, equal opportunity fully discounts the larger book. Targeting $S_a/(S_a + S_b)$ instead destroys fairness on gains (KS gap 0.233); a $\sqrt{\cdot}$ compromise sits at 0.051. That choice is a compensation-philosophy decision, not a modelling one, and should be made explicitly.

6 Contestability

Two components, multiplied.

Stake. Kalai–Smorodinsky equalises $g_a/G_a^{\max} = g_b/G_b^{\max}$, so misassigning zip z shifts that balance by

$$\text{stake}_z = \frac{u_a(z)}{G_a^{\max}} + \frac{u_b(z)}{G_b^{\max}}, \tag{6}$$

which ranged over $[0.0138,\ 0.1464]$, with the top five zips carrying 26% of the total.

Doubt. The flip rate $2\min(p, 1-p)$ under uncertainty. Sweeping $\lambda \in [0.10, 0.50]$ and $\theta \in [0.20, 0.60]$ moved only five zips; perturbing the *data* by 10% on sales and 6% on opportunity moved thirty-six.

> Parameter uncertainty is nearly irrelevant on this instance; data uncertainty dominates. Getting A_z and B_z right matters more than settling λ.

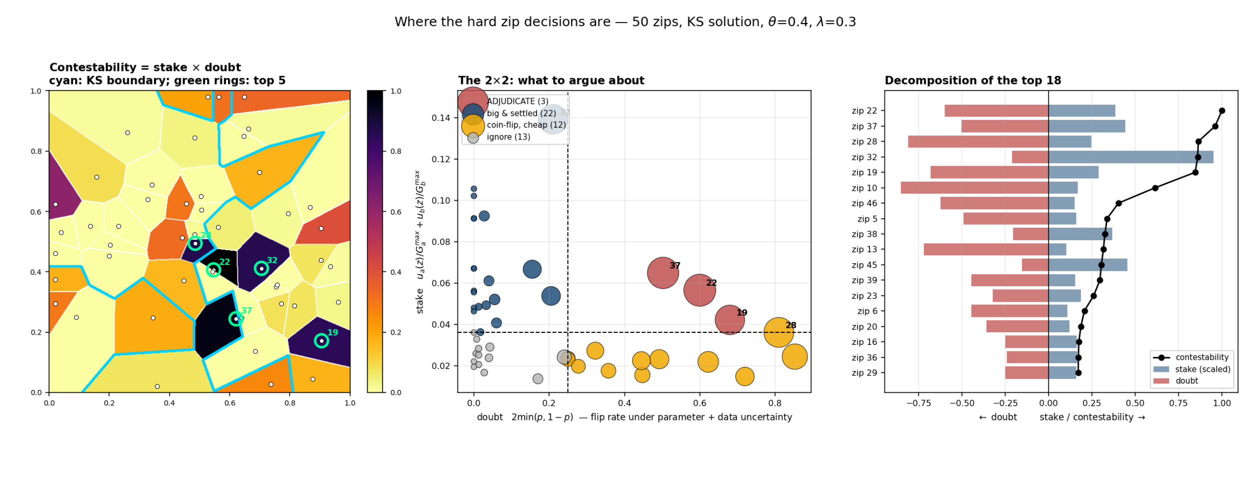

Figure 3: Contestability as stake × doubt. Left: mapped, with the allocation boundary in cyan and the five most contested zips ringed. Centre: the 2 × 2. Right: decomposition of the top eighteen.

Only three zips fall in the adjudicate quadrant. Zip 32 is the instructive near-miss: it carries the largest stake in the instance (0.139, more than double the next) but doubt of only 0.21. It will *look* like the fight, being the largest zip near the boundary, while the model is in fact confident about it. The real contests are zips 22, 37 and 19, each about a third its size. Contestability is concentrated: the top five zips account for half the total.

7 Exact parameter breakpoints

Write $P_z = A_z + \theta B_z$ and $Q_z = B_z + \theta A_z$, so that $u_a(z) = (1-\lambda)P_z + \lambda M_z$. Zips i and j exchange places in the ratio order exactly when

$$(1-\lambda)K_{ij} + \lambda L_{ij} = 0 \quad \Longrightarrow \quad \lambda^\star_{ij} = \frac{K_{ij}}{K_{ij} - L_{ij}}, \tag{7}$$

$$K_{ij} = P_i Q_j - P_j Q_i, \qquad L_{ij} = P_i M_j - P_j M_i + Q_j M_i - Q_i M_j. \tag{8}$$

One explicit expression per pair. In θ the same condition is quadratic, hence also closed form.

On the instance there are 88 such swaps in $(0,1)$, but only 13 distinct allocations across $\lambda \in [0.02, 0.90]$: most swaps occur between zips already on the same side of the cut. Observed breakpoints matched the analytic prediction to about 10^{-4}.

$\lambda = 0.30$ sits inside a flat interval running from 0.205 to 0.505. Anyone who believes a dollar of untapped opportunity is worth between twenty and fifty cents of booked production obtains the same map. Nearly all sensitivity is crowded below $\lambda \approx 0.11$.

This is a stronger instrument than a local derivative. Rather than reporting $dk/d\lambda$, one can state the complete set of possible answers and exactly which belief produces each.

8 Which criterion to use

Recommendation: Nash, with the egalitarian criterion as fallback below λ^*.

Three properties separate Nash from the alternatives once the problem is discrete.

It is exactly solvable. By Proposition 1 the Nash optimum is a prefix, so enumerating $n+1$ options after one sort gives the exact answer. The equalisation criteria do not inherit that structure and need search for their exact optima.

It cannot buy fairness by destroying value. Maximising a product is Pareto efficient by construction: no allocation makes both wholesalers better off. The min-|difference| criteria carry no such guarantee, and the unconstrained MILP of Section 9 demonstrates the failure — a gap of 0.000161 bought at 82% of attainable welfare. On the clean instance all four criteria happen to land on the frontier, but only Nash guarantees it.

It delivers EF1. Neither wholesaler prefers the other's territory after removing a single zip. That is the strongest fairness guarantee available for indivisible goods and it is legible in a room.

Against Kalai–Smorodinsky. An earlier draft of this work preferred KS on the strength of its closed-form comparative statics, which require only densities rather than density derivatives. That argument is a *continuum* argument and does not survive the move to discrete data, where there are no densities. What remains is KS's distinctive feature, equalising fractions of attainable gain. Note the direction in which that cuts: $G_b^{\max}$ exceeds $G_a^{\max}$ precisely because b brought the smaller book and therefore has the lower baseline to clear, so KS systematically awards *more absolute gain to the wholesaler who brought less production.* Defensible in principle; a difficult sentence to deliver to a.

The one thing Nash cannot do. It requires $g_a, g_b > 0$, so below the feasibility threshold λ^* the bargaining set is empty and the criterion is undefined. Report λ^* explicitly; on the instances tested it was 0.069 under the net convention, well below any defensible λ. If the data place you below it, the merger is negative-sum for the wholesalers and the conversation is not about which criterion to use.

If seniority should count. Use asymmetric Nash, $\max g_a^{\omega} g_b^{1-\omega}$, which retains the prefix property with threshold $\frac{1-\omega}{\omega} \cdot \frac{g_a}{g_b}$. Making the weight explicit is preferable to letting a criterion encode a tilt nobody selected.

Report all of them regardless. The prefix table yields every criterion at no extra cost, and the comparison is itself evidence: agreement across criteria is a far stronger claim than any single one.

9 Contiguity, and the districting ILP

Nothing above requires each wholesaler's territory to be geographically connected, and in general it is not: on the worked instance the Kalai–Smorodinsky allocation left a in four pieces and b in three. Imposing connectivity turns the problem into a *districting* integer program — the formulation used for electoral redistricting and sales-territory alignment. With two wholesalers it has one binary per zip.

9.1 Formulation

$$\begin{aligned} &\text{decision} && x_z \in \{0,1\}, \quad 1 \text{ if } z \text{ goes to } a, && (9)\\ &\text{fairness} && g_a = \textstyle\sum_z x_z u_a(z) - S_a, \qquad g_b = \textstyle\sum_z (1-x_z) u_b(z) - S_b, && (10)\\ &\text{objective} && \min \quad t + \rho \textstyle\sum_{e\in E} y_e, \qquad t \;\geq\; \pm\Big(\frac{g_a}{G_a^{\max}} - \frac{g_b}{G_b^{\max}}\Big), \qquad y_e \;\geq\; |x_u - x_v|. && (11) \end{aligned}$$

Both gains are linear in x, so the fairness gap linearises with one auxiliary variable. The variables y_e count *boundary edges* — the perimeter of the partition — and ρ prices compactness.

The compactness term is not optional. Fairness alone is degenerate: with fine-grained values, many allocations achieve essentially the same gap, so contiguity cuts never bite — the solver simply returns another equally fair, equally disconnected allocation. A first implementation without the ρ term thrashed for seventeen rounds without converging. With $\rho = 2 \times 10^{-3}$ the same instance solved in **two rounds and 4.5 seconds**.

9.2 Contiguity by separator cuts

Fix a root zip for each side. Solve; if a's territory splits, then for each component S not containing a's root add

$$|S| \sum_{w \in N(S)} x_w \quad \geq \sum_{v \in S} x_v, \tag{12}$$

"if any of S is with a, some neighbour just outside S must be too"; symmetric in $(1-x)$ for b. Re-solve and repeat. The alternative is a single-commodity *flow* formulation — ship one unit from every assigned zip to its root along in-district edges — which is exact in one solve but adds $O(|E|)$ continuous variables per district. Cut generation is preferable at ZCTA scale, and aggregating one cut per island rather than one per zip matters for convergence.

9.3 The price of contiguity, and a second trap

	KS gap	welfare	% of max	pieces a/b
unconstrained prefix	0.022398	28.301	85.8%	4/3
greedy island repair	0.415616	—	—	1/1
MILP, no welfare floor	0.000161	27.094	82.1%	1/1
MILP, welfare floor 0.85	0.001604	28.055	85.1%	1/1

Two readings. First, the greedy repair heuristic is not merely suboptimal but badly so — more than two thousand times the exact gap — and should be used only to detect fragmentation, never to fix it.

Second, and less obvious: the unconstrained MILP attains a very small gap partly by making *both* wholesalers worse off, giving up 18% of attainable welfare. This is the same pathology as the prefix violation above — minimising a difference is not a monotone objective. The remedy is a welfare floor,

$$g_a + g_b \;\geq\; \kappa \cdot \max_S \big(g_a + g_b\big), \tag{13}$$

whose right-hand side is available in closed form because the utilitarian optimum is the prefix $\{z : u_a(z) > u_b(z)\}$. At $\kappa = 0.85$ the solver returns a connected allocation with gap 0.0016 at 85.1% of maximum welfare — fourteen times fairer than the best prefix at essentially the same welfare.

> Contiguity is close to free when solved properly. The earlier impression that it costs 8–13% of achievable fairness was an artefact of the greedy heuristic, not a property of the problem, and it removes the motivation for heavier machinery such as semi-discrete optimal transport.

9.4 Contiguous Nash on the worked instance

Applying the same machinery with the Nash objective requires care: the product is not linear, and a weighted-sum scalarisation $wg_a + (1-w)g_b$ collapses to corner solutions when the frontier is close to linear, which it is here. An ε-constraint sweep — maximise g_a subject to $g_b \geq \tau$, varying τ — traces the frontier properly and the Nash point is the sweep member with the largest product.

	g_a	g_b	$g_a g_b$	perimeter	pieces a/b
Nash, unconstrained	4.9360	4.8807	24.0910	—	2/3
Nash, contiguous (MILP)	5.1048	4.7082	24.0348	19	1/1

The price of contiguity under Nash is 0.23% of the bargaining product. Fragmentation in the unconstrained solution is minor — single-zip islands — and removing it costs almost nothing.

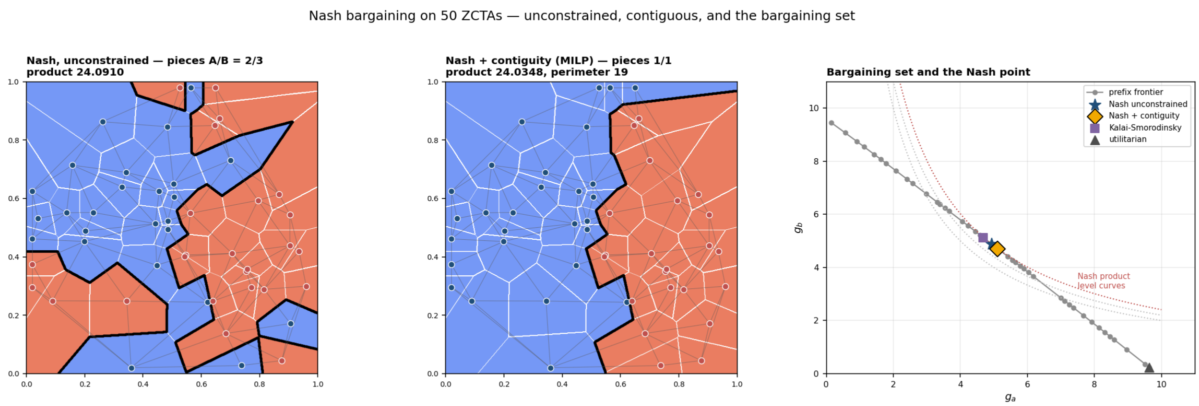

Figure 4: Left: the unconstrained Nash allocation, with the Rook adjacency graph overlaid; each wholesaler holds a stray single-zip island. Centre: the contiguous MILP solution, one connected piece each. Right: the bargaining set. Grey points are the prefix frontier; dotted curves are level sets of $g_a g_b$; the Nash point is where the highest attainable level curve touches the frontier. Kalai–Smorodinsky sits adjacent to it, and the utilitarian solution sits at the far corner where b is left with almost nothing.

The right-hand panel is the clearest single picture of what the exercise does. The frontier is nearly linear, which is why criteria agree so often; the utilitarian corner shows what happens without a fairness objective at all.

9.5 Implementation

The accompanying `territory.py` operates directly on a `networkx` graph of ZCTAs with Rook adjacency and attributes `rep_a`, `rep_b`, `A`, `B`, `M`, `state`: it validates the headroom condition, builds the bipartite overlap graph of wholesaler pairs and reports its component structure, solves each pair by prefix table, and writes results back onto the graph for mapping. `districting.py` adds the contiguous MILP above, with `solve_contiguous_nash` performing the ε-constraint sweep. Two practical notes: ρ needs tuning to scale — too small and cut generation thrashes, too large and it distorts fairness — and at national scale a solver with lazy-constraint callbacks is preferable to re-solving from scratch each round.

Setting `respect_state=True` deletes cross-state edges before enforcing connectivity. For life and annuity distribution this may be the binding constraint rather than contiguity, since product availability and producer appointments are state-scoped; it should be settled with distribution before parameters are tuned.

10 Scope

Covered. Two wholesalers, discrete postal codes, arbitrary sales and opportunity data, five selection criteria, exact parameter sensitivity, contestability triage.

Not covered. Capacity and travel constraints. Three or more wholesalers, for which no threshold rule exists and the natural extension is leximin over each dense component of the overlap graph. Estimation of θ from historical reassignments, on which everything ultimately rests.

Immediate checks on real data.

1. $\mathrm{corr}(A_z, B_z)$ across zips, which determines whether the two books overlap or separate and hence how contested the exercise is.
2. Connected components of the allocation on the true adjacency graph, which prices contiguity rather than assuming it.
3. The measurement error on A_z, B_z, M_z, since data uncertainty rather than parameter uncertainty sets which zips require adjudication.